Dosage Form and Routes of Administration

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Lecture Outline

- Introduction
- Oral
  - Solid, liquid
- Parenteral
  - Intravenous, intramuscular, subcutaneous
- Body cavity-specific
  - Eye, ear, nose, mouth, lung, rectum, vagina
- Topical
  - Cream, ointment, lotion, liniment, transdermal patch

Definition and Types

- Dosage form:
  - The final physical form of a drug, containing the active pharmaceutical ingredient(s) in which measurable doses can be administered to a patient.

Types:

1. Oral
2. Parenteral
3. Body cavity-specific
4. Topical

Oral dosage forms

Solid and liquid preparations taken by mouth

Oral Dosage Forms

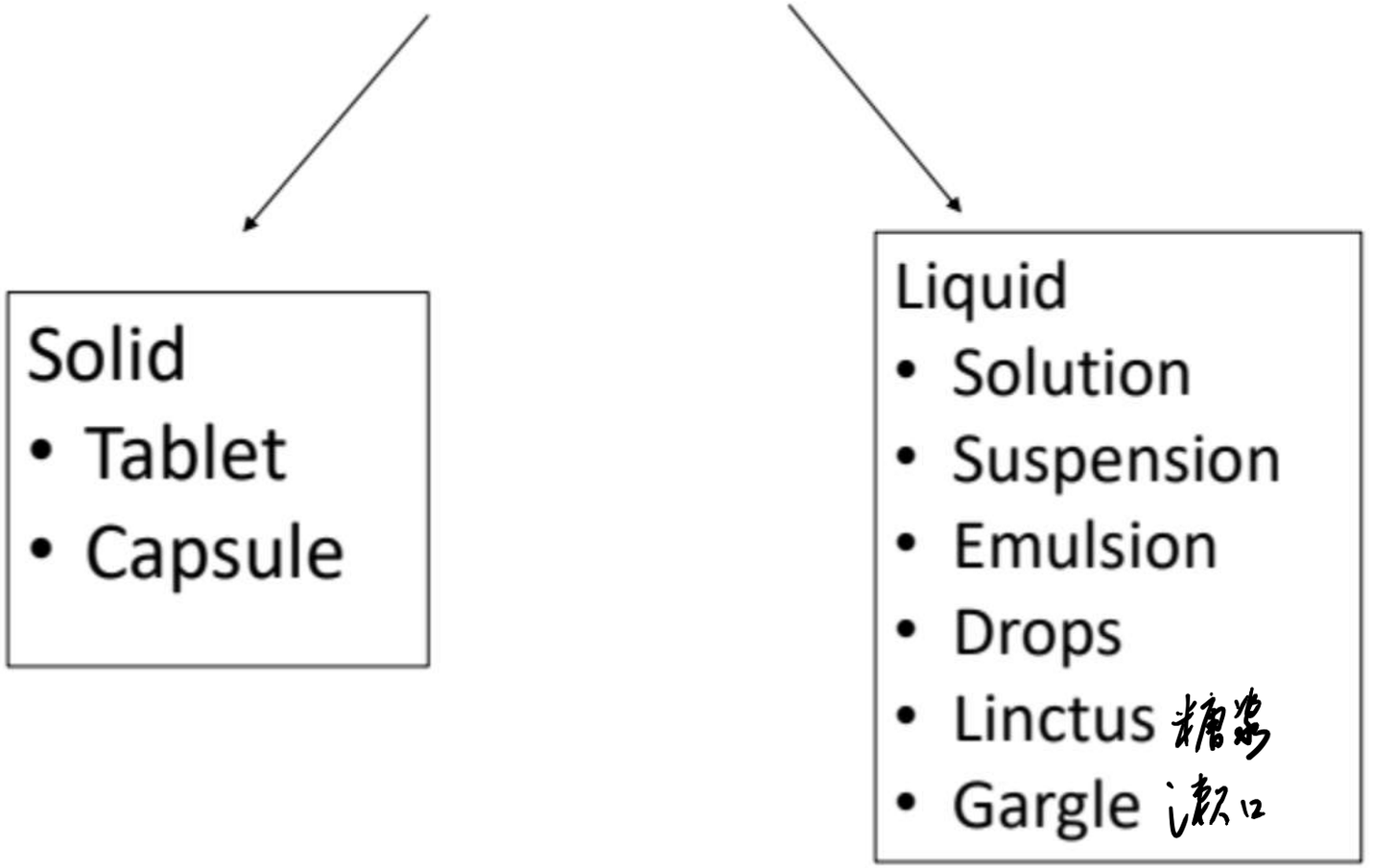
- Intended for administration through the mouth





Oral Dosage Forms

- Intended for administration through the mouth



Oral Dosage Forms - Solid

- Tablet



- Capsule



Oral Dosage Forms -Solid

- Tablet

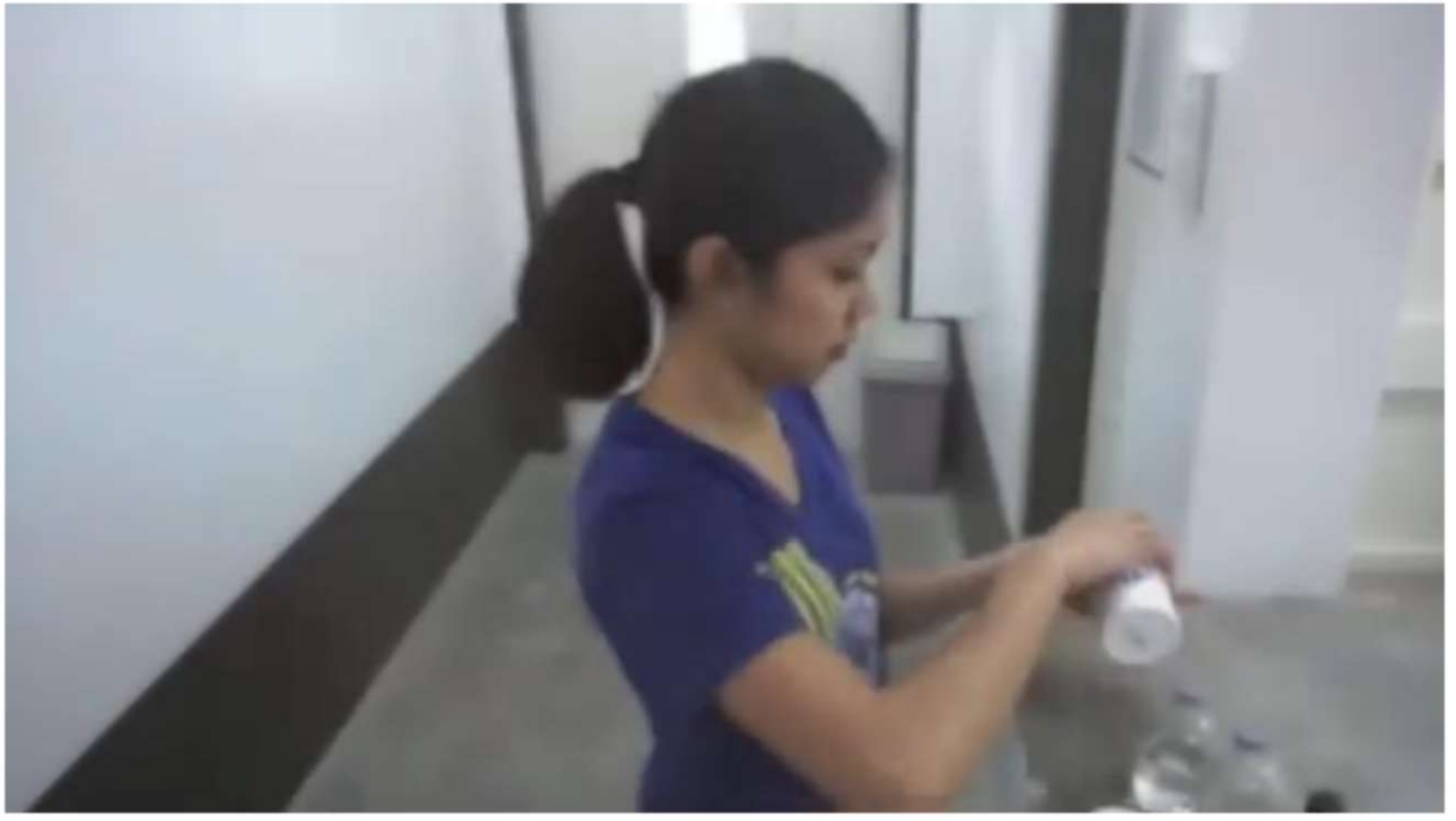
- Drug in powder form compressed



- Modification
  - Sublingual
  - Chewable
  - Soluble/effervescence
  - Lozenges
  - Slow/sustained/controlled release
  - Enteric-coated

Sublingual

- Video on sublingual tablet how to properly use sublingual tablet



Tablet: Modification

- Sublingual (under the tongue)

Ordinary Tablet

Tablet swallowed

↓

Dissolved in the stomach

↓

Liver (first-pass metabolism)

↓

Heart

↓

Blood stream



Sublingual Tablet

Tablet hold under the tongue

↓

Blood stream

Advantage
-allow rapid onset of effect

Example: nitroglycerin

- Nitroglycerin, also known as glyceryl trinitrate (GTN) is used to relieve chest pain (angina) from myocardial infarction (heart attack)

- Sublingual tablet is used for rapid effect.

Tablet- Modification



- Chewable
  - For children or those who cannot swallow whole tablet.

- Effervescence
  - faster onset of action and enhanced absorption compared to standard tablets
  - ideal for rapid relief and for individuals with swallowing difficulties, such as children and the elderly
  - Improve compliance.



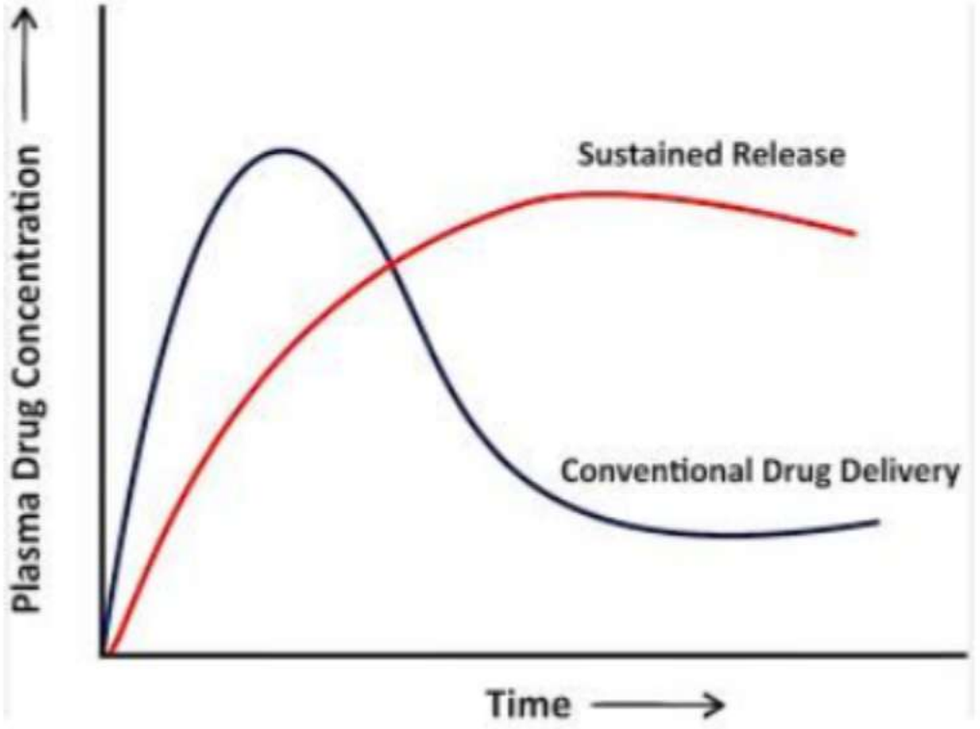


- Lozenges 含片
  - Contain medicament
  - Suck in the mouth to medicate mouth & throat



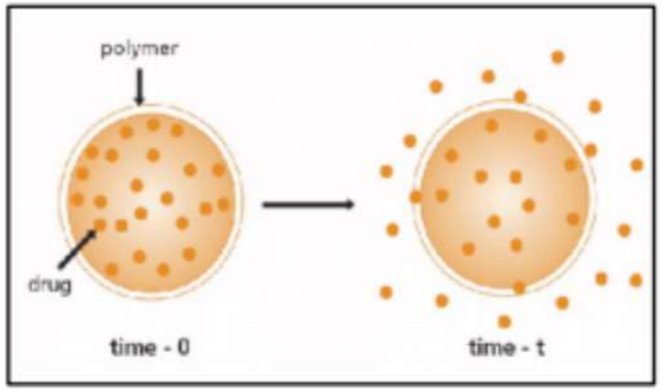
Tablet - Modification

- Slow/sustained/controlled release
  - Medication is released slowly for a prolonged effect.
  - Ordinary/conventional tablets break up in the stomach and absorbed in the stomach at once.



Do not chew the tablet

Tablet - Modification



- Slow/sustained/controlled release breaks up slowly
- Medication is dispersed in the polymer matrix

Mechanism of Release

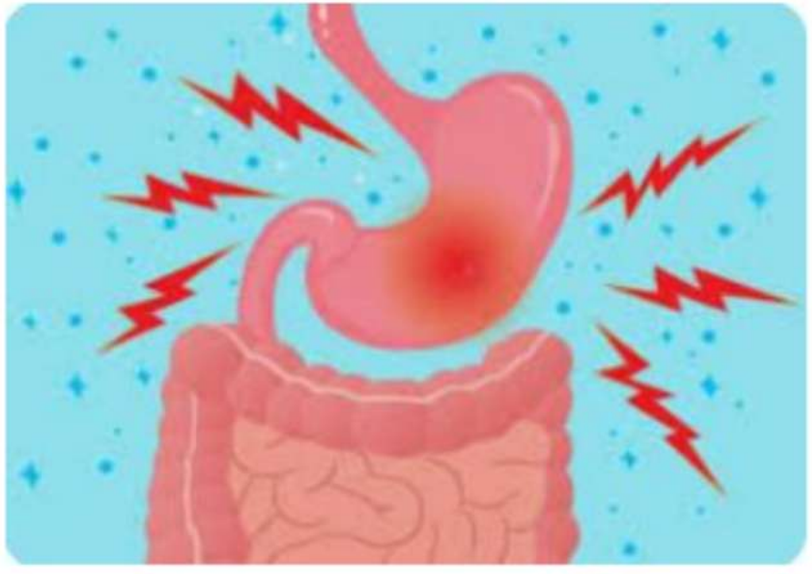
The dissolved drug molecules move through the gel layer (hydrophilic matrix) or through water-filled pores (hydrophobic matrix) down a concentration gradient.

Advantages

- Reduces the dosing frequency (improve adherence)
- Minimise side effect (High peak concentrations of certain medication can sometimes lead to adverse side effects)

Tablets- Modification

If the drug in a tablet causes irritation of the stomach wall. Can you modify the tablet to prevent this problem?

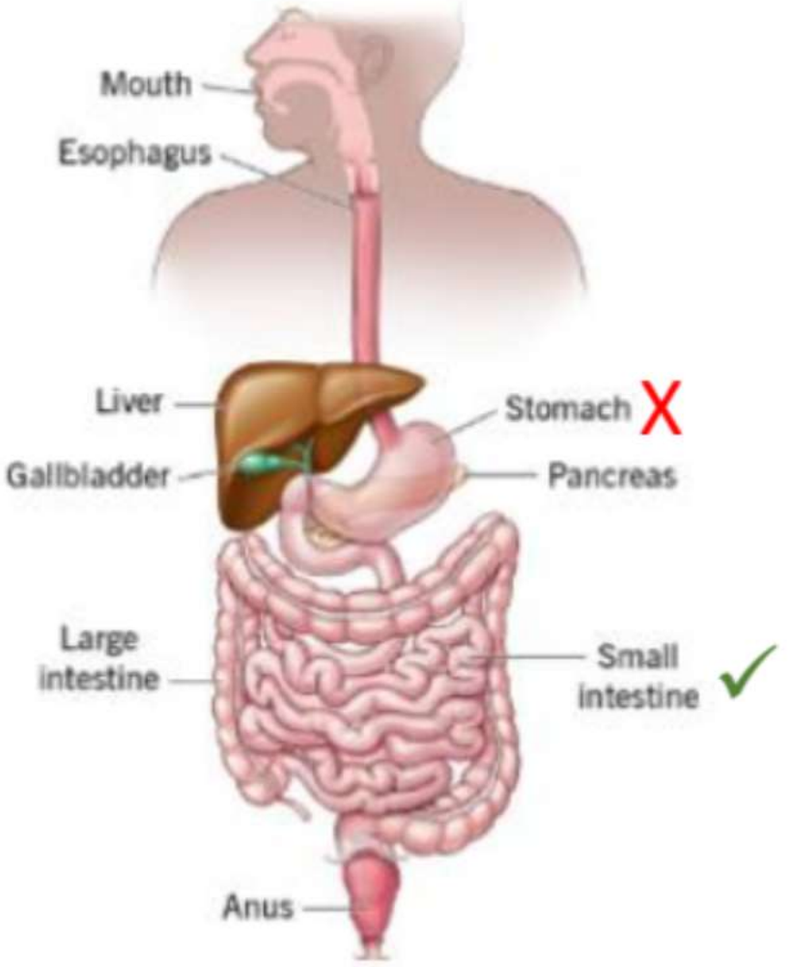




Tablets- Modification

肠溶

- Enteric coated tablets
  - Tablet coating that dissolves at higher pH
  - Enteric coated tablet does not dissolve in acid environment of stomach. Only dissolves in the small intestine.
  - Swallow without chewing the tablet
  - Protect drug from dissolved by the acid environment.
  - Protect drug from degradation by acid
  - Protect stomach mucosa from the drug



Advantages of tablet

- Portable, easy to carry
- Precision of dose
- Palatable 适口性
- Stable and have long expiry date
- Low cost as can manufacture rapidly on large scale.

Solid form - Capsule

- medication in solid (powder), semi-solid or gel form is encapsulated in a gelatin gel.
- Advantages
  - Palatable- masks the unpleasant taste with shells
  - Formulation Versatility: Capsules can hold a wide variety of fillings e.g oil, semi-solid, powder
  - Portable
  - Gelatin shell allows easy release of the drug- faster absorption
  - Easy identify (colour and wording)
- Disadvantages
  - Relative costly: filling capsules takes time.



Single Answer Question

Which tablet modification is designed to bypass stomach acid?

A. Chewable
B. Sublingual
C. Enteric-coated
D. Effervescent

Single Answer Question

Which tablet modification is designed to bypass stomach acid?

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Oral Dosage Forms - Liquid

- One or more active ingredients in a suitable liquid vehicle
- Most common liquid dose form. May contain:
  - Emulsifying, suspending, thickening agents
  - Antimicrobial agents
  - Sweetening, flavouring or colouring agents

- Oral solution, suspension, emulsion, drops, linctus

Oral Liquid Dosage Forms - Solution

- Oral solution - A clear liquid preparation containing drug particles (one or more ingredients) dissolved in a suitable vehicle.



Oral Liquid Dosage Forms - Suspension

- Suspension – containing drug particles that do not dissolve but suspended in a suitable vehicle. It can form sediment. Needed to shake the bottle to redisperse the drug particles for pouring out the suitable dosage of drugs.

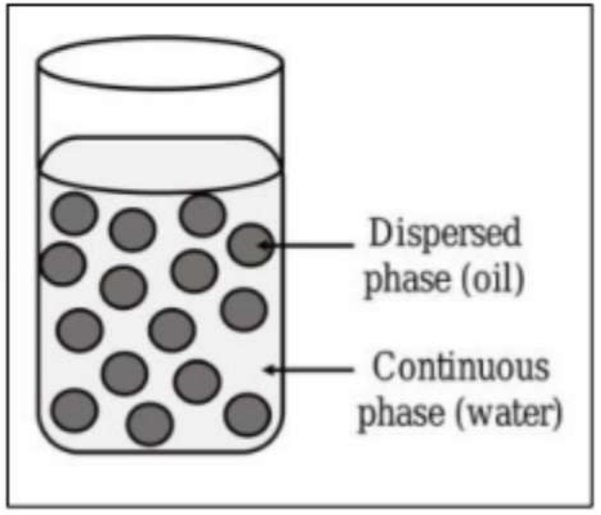




Oral Liquid Dosage Forms - Emulsion

Emulsion – A mixture of two immiscible liquids where droplets of one liquid is dispersed into the other

oil in water E.g. Cod liver oil





Oral Liquid Dosage Forms - Emulsion

Advantages of Emulsion

- Increase the solubility, for drugs that only dissolve in oil.
- Improve the palatability (taste/texture masking) of the medication as the bad taste of oil is masked by flavouring agents in the aqueous phase
- protect the active ingredients from degradation in the gastrointestinal (GI) tract.

Oral Liquid Dosage Forms

Suspension & Emulsion

- Before pouring, “SHAKE THE BOTTLE”

so the drug remains in uniform suspension (i.e. well mixed) during pouring.



Oral Liquid Dosage Forms - Drops

- Drops – Drug in liquid form which is concentrated, allowing use of a few drops per dose. Not for patient to swallow a large volume.
  - Used for dosing young children

e.g. multivitamin oral drops for infant

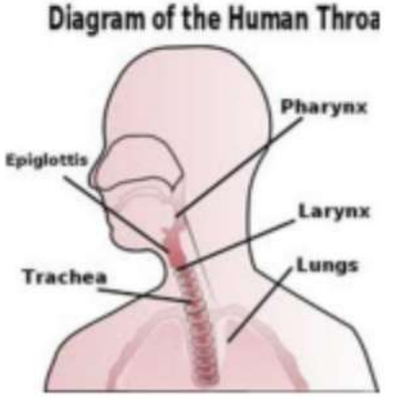


Oral Liquid Dosage Forms - Linctus



Linctus

- a viscous oral liquid that contains drug particles that is dissolved in a suitable base and has a high concentration of sugar.
- the viscous (thick) liquid gives a demulcent (soothing) effect when swallowed slowly.
- Commonly used in cough medicines.
- cough syrup – to give a soothing effect by coating the throat



Liquid oral forms compared to solid oral forms

Advantages

- Easier to swallow especially for children
- Can add different colours and different flavours.
- Drugs in solution are generally absorbed faster

Disadvantages

- Less portable
- Less accurate dosing - When pouring out, can have slight variation of dose .
- Some drugs are less stable in liquid form
- After some time, potential microbial contamination

Parenteral dosage forms

Medicines administered by injection or infusion, bypassing the gastrointestinal tract

B. Parenteral Dosage Forms



Parenteral Dosage Forms

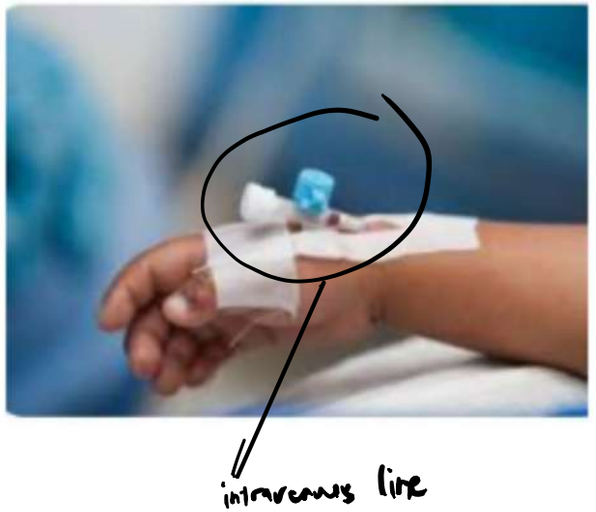
Parenteral:

- Not by the gastrointestinal tract
  - BY INJECTION (through vein, muscle, skin etc)
- Preparation needs to be sterile and particle-free
- Reasons to inject:
  - Patient is unconscious/uncooperative
  - No oral formulation available (e.g. insulin).
  - Need a rapid onset
  - Need high levels of drug in the blood.

Parenteral Dosage Forms

- Common parenteral routes:
  - Intravenous (I.V)
  - Intramuscuslar (I.M)
  - Subcutaneous (SC)

Parenteral Dosage Forms



- Intravenous
  - Into a vein
  - Effect is most rapid
  - Usual volume: 1 - 15mL, or infusion



- Intramuscular
  - Into muscle tissue, usually at arm, thigh, buttock
  - Usually 1-2mL (larger volume causes pain)



Parenteral Dosage Forms

- Subcutaneous
  - Under the skin (dermis)
    - E.g. insulin, local anaesthetic.
  - Normally < 1 mL

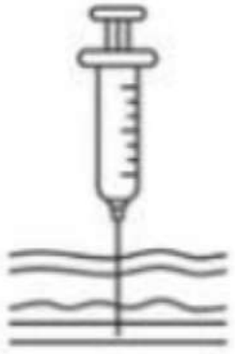


Parenteral Dosage Forms

- Video on how to administer subcutaneous insulin injection



Parenteral Dosage Forms



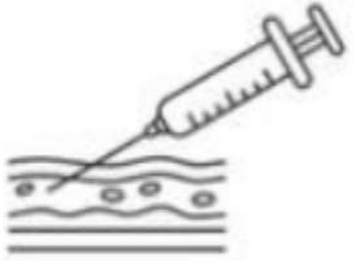
IM

INTRAMUSCULAR



SC

SUBCUTANEOUS



IV

INTRAVENOUS

Parenteral routes

Intravenous (IV)

Into a vein

Immediate systemic availability

Used when rapid action or controlled infusion is required

Intramuscular (IM)

Into skeletal muscle

Absorption varies with formulation and blood flow

Used for selected vaccines, medicines etc

Subcutaneous (SC)

Into tissue beneath the skin

Often suitable for smaller volumes

Used for medicines such as insulin and some anticoagulants

Single answer question

Which statement best describes the IV route?

A. Slow effect, given under the skin
B. Rapid effect, given directly into a vein
C. Used only for children
D. More painful than IM injection

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Body-cavity-specific dosage forms

Preparations designed for local treatment or, in some cases, systemic absorption

C. Body-Cavity Specific Dosage Forms

Body-Cavity Specific Dose Forms

- Body cavity-specific dosage forms
- Administer the drug at local specific cavity of the body
  - Eye, ear, nose, mouth, lung, rectum, vagina

Body-Cavity Specific Dosage Forms



- Eye
  - Eye drop
    - A sterile solution for administering drug into the eye
    - Usual medications include antimicrobials, eye lubricant, mydriatics (cause the pupil of the eye to dilate (widen)), miotics (constrict the pupil and are primarily used to treat glaucoma) E.g. Eye infection use eye drop as a local antibiotic.
    - Dispose bottle ~ one month after opening.

Body-Cavity Specific Dosage Forms

Eye ointment

- Medication in smooth oily base, for eye lid or absorption into eye
- Gives prolonged action. Can give a longer effect than eye drops



Body-Cavity Specific Dosage Forms

- Ear
  - Ear drop
    - Liquid solution to be admistered into the ear
    - E.g antibiotic for ear infection, cerumol (dissolve ear wax)



Body-Cavity Specific Dosage Forms

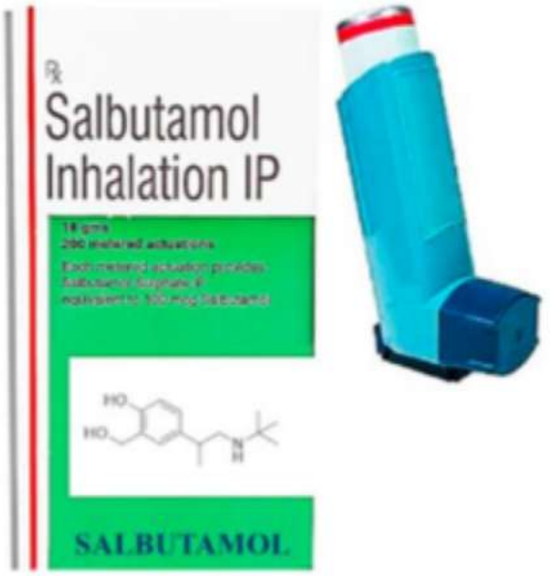


- Nose
  - Nasal drops or nasal spray
    - Administering drug for local action for nasal passage
    - Commonly used for nasal congestion.

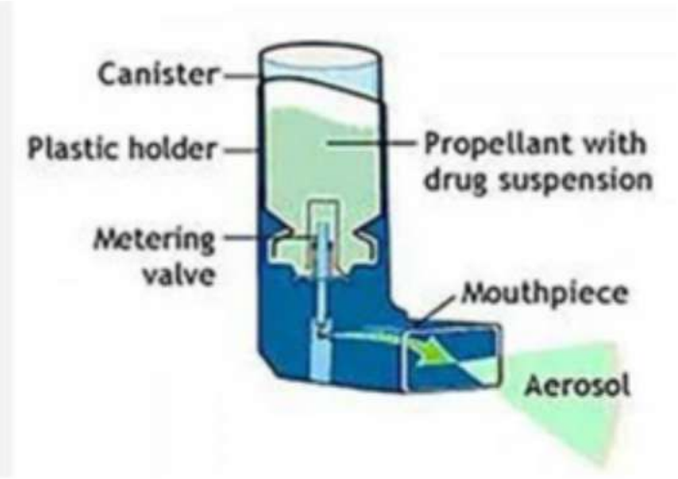
- Mouth
  - Mouth wash or gargle
    - Not for swallowing but gargling for pain in mouth or sore throat
  - Mouth gel
    - For painful mouth ulcer e.g. Bonjela gel

Body-Cavity Specific Dosage Forms

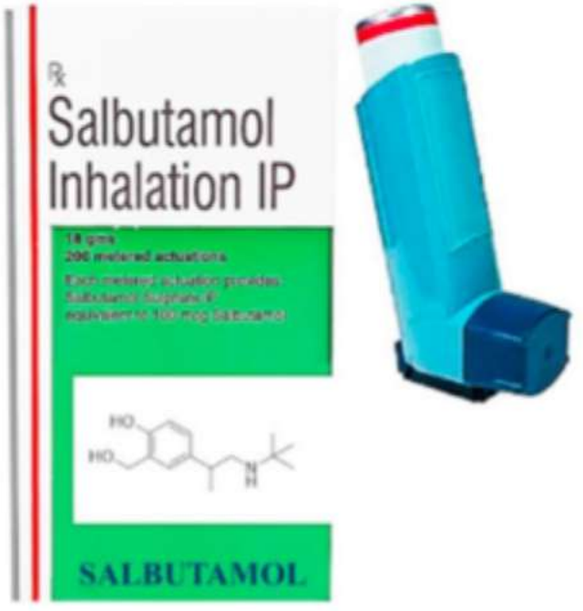
- Lung
  - Administer the drug to the lung.
  - For example for bronchial asthma treatment.



Body-Cavity Specific Dosage Forms



- Metered dose inhalers
  - a device used to deliver a specific dose of medication directly into the respiratory tract leading to the lungs
  - Pharmacaeutical aerosol (fine particles of drug or medicine (either solid or liquid, or both) that is kept in a container under pressure and that is released as a fine spray when a button is pressed.



MDI

Body-Cavity Specific Dosage Forms

- Video on how to use MDI.



Body-Cavity Specific Dosage Forms

- Advantages of MDI
  - It is easy to carry around.
  - It can deliver multiple doses.
  - There is a low risk of bacterial contamination.
- Disadvantages of MDI
  - The need for correct, coordinated use of the inhaler.
  - The potential for the medication to irritate the mouth and throat.

Body-Cavity Specific Dosage Forms

- Dry powder inhalers (DPIs)
  - an instrument that administers dry powder medication to the lungs.
  - **Breath-actuated:** The medication is released when the patient inhales, which removes the need to coordinate breathing with actuating a device.



DPI

Question

A 28-year-old woman suffered from acute shortness of breath caused by asthma at home. She needs a drug that reaches the lungs quickly and acts within minutes. Which route of administration is the most appropriate?

A. Oral tablet

B. Intramuscular injection

C. Metered-dose inhaler

D. Oral syrup

Question

A 28-year-old woman presents with acute chest tightness caused by asthma. She needs a drug that reaches the lungs quickly and acts within minutes. Which route of administration is the most appropriate?

A. Oral tablet

B. Intramuscular injection

C. Metered-dose inhaler

D. Oral syrup

Body-Cavity Specific Dosage Forms

- Rectum
  - Insertion into anal canal.
  - Examples suppository or liquid form

    E.g suppository is in a

    bullet shape
    - the drug will melt in the body temperature to release the medication.
  - Can be a local effect in the rectal musosa or for systemic absorption



Example: Rectal Paracetamol

Commonly prescribed for children primarily when they **cannot take oral medication** due to

- Uncooperative
- Frequent vomiting
- Seizure





Body-Cavity Specific Dosage Forms

- Rectum
  - Enema
    - A liquid form push into the anal canal
    - To promote the bowel movement for constipation or to empty the bowel for certain medical procedures.



Body-Cavity Specific Dosage Forms

- Vagina

Drugs that are inserted into vagina

- Cream, gel, suppository, ring, tablet
- Example: Clotrimazole tablet as an antifungal medication.



Single Correct Answer Questions

Which route is most suitable for children with frequent vomiting who requires paracetamol?

A. Oral tablet

B. Sublingual

C. Rectal suppository

D. Transdermal patch

Single Correct Answer Questions

Which route is most suitable for children with frequent vomiting who require paracetamol?

A. Oral tablet

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D. Transdermal patch

Topical dosage forms

Preparations applied to the skin or other accessible surfaces

D. Topical Dosage Forms

Topical Dosage Forms

- Pertaining to a particular surface area / applied to the skin
  - Cream
  - Gel
  - Ointment
  - Lotion
  - Liniment
  - Transdermal patch

Topical Dosage Forms



- Cream
  - A semi-solid emulsion (aqueous or oily) for topical use)
  - Commonly used for anti-inflammatory, antiseptic

- Gel
  - A transparent or translucent semi solid preparation
  - Liquifies upon contact with skin.
  - Commonly used for anti-inflammatory, antiseptic

Topical Dosage Forms



- Ointment
  - A semi-solid preparation in a stiff base
  - Greasy, emollient (softens the skin)
  - Occlusive (increase the skin moisture by reducing water loss and increase drug absorption).
  - Used as base to deliver medication e.g. Tiger Balm® ointment.

- Lotion
  - A liquid suspension preparation, give a cooling effect to the skin by evaporation of the solvent
  - E.g. calamine lotion for skin irritation and itching

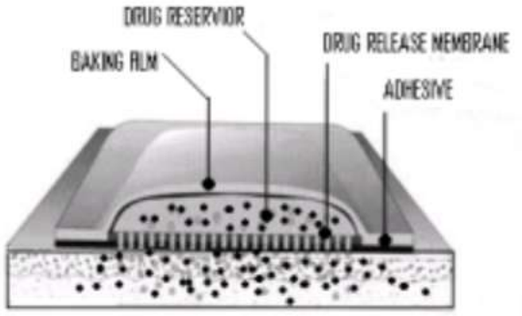


Topical Dosage Forms



- Liniment
  - A liquid or semi-liquid
  - Applied to the skin by rubbing.
  - contain rubefacient such as methyl salicyclate
  - Rubefacient produces redness of the skin by causing dilation of the capillaries and an increase in blood circulation and cause the skin to feel warm.
  - This sensation is interrupted the pain signal underlying muscle or joint- intended to reduce pain. Also know as counter irritant effect.

Topical Dosage Forms



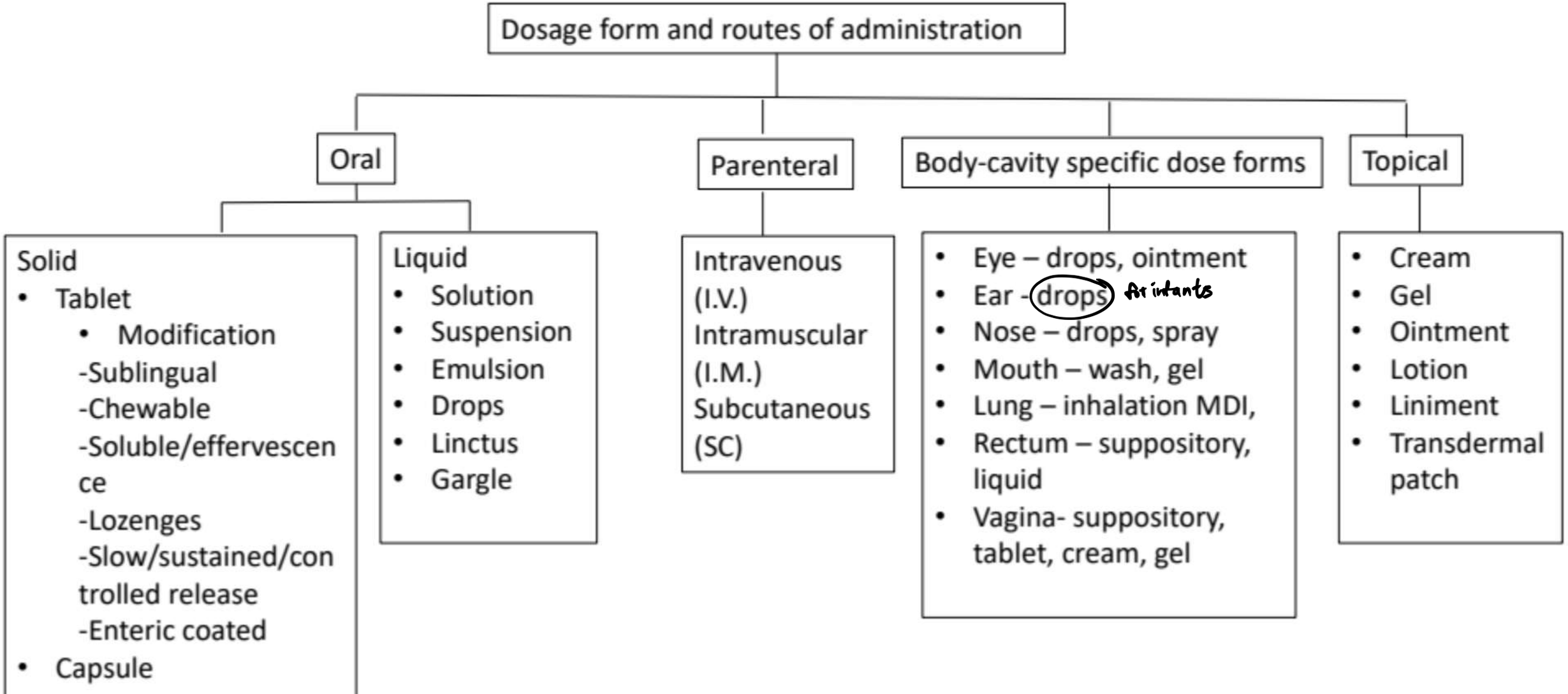
- Transdermal patch
  - a sheet-like device (patch) that adheres to the skin and is loaded with drug molecules, allowing the drug to diffuse through the skin into the bloodstream, thereby exerting therapeutic effects throughout the body.
  - E.g smoking cessation (nicotine), pain relief (fentanyl, lidocaine), hormone replacement (estrogen, testosterone)
  - Advantages
    - bypasses the first pass effect in liver.
    - reduced dosing frequency by releasing medication over a longer period of time.



Routes of administration, bioavailability and general characteristics

Route	Bioavailability	Characteristic
Intravenous (IV)	100 (by definition)	Most rapid onset
Intramuscular (IM)	75 to ≤ 100	Larger volume often feasible: may be painful
Subcutaneous	75 to ≤ 100	Smaller volumes than IM: may be painful
Oral (PO)	5 to < 100	Most convenient; First pass effect may be important
Rectal (PR)	30 to < 100	Less first-pass effect than oral
Inhalation	5 to < 100	Often very rapid onset
Transdermal	80 to ≤ 100	Usually very slow absorption; used for lack of first-pass effect: prolonged duration of action

Summary


Dosage form and routes of administration
Oral
Parenteral
Body-cavity specific dose forms
Topical
Solid
• Tablet
• Modification
-Sublingual
-Chewable
-Soluble/effervescence
-Lozenges
-Slow/sustained/controlled release
-Enteric coated
• Capsule
Liquid
• Solution
• Suspension
• Emulsion
• Drops
• Linctus
• Gargle
Intravenous (I.V.)
Intramuscular (I.M.)
Subcutaneous (SC)
• Eye – drops, ointment
• Ear - drops
• Nose – drops, spray
• Mouth – wash, gel
• Lung – inhalation MDI,
• Rectum – suppository, liquid
• Vagina- suppository, tablet, cream, gel
• Cream
• Gel
• Ointment
• Lotion
• Liniment
• Transdermal patch

Thank you

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